

universität freiburg



CS

DEPARTMENT OF
COMPUTER SCIENCE

Test and Reliability

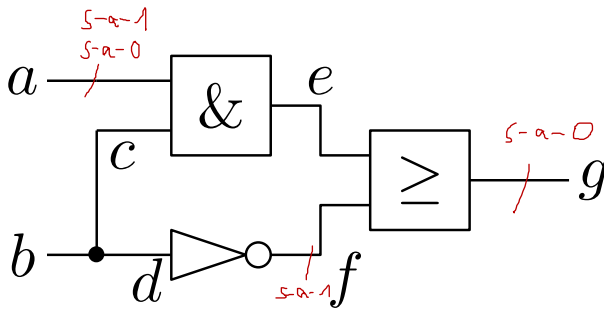
Exercise Sheet 3

Prof. Dr. Ralf Wimmer
Dr. Tobias Seufert

June 6, 2024



Exercise 1) *Fault* Parallel Simulation with X



4 faults

+ 1 fault free = 5 simulations
each 3-valued = 2 bits per value
 $\Rightarrow$ in total $5 * 2 = 10$ bit, 16 bits avail.
 $\Rightarrow$ can do in 1 simulation pass

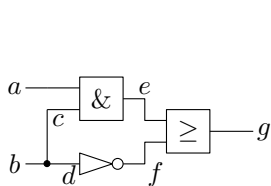
Exercise 1a) Fault Parallel Simulation with X

0 = 00

1 = 11

X = 01

dc = don't care

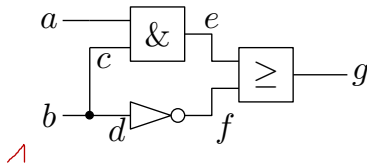


	ff	a/0	a/1	f/1	g/0	dc	dc	dc
I_a	00	11	11	00	00	00	00	00
S_a	00	00	11	00	00	00	00	00
I_f	00	00	00	11	00	00	00	00
S_f	00	00	00	11	00	00	00	00
I_g	00	00	00	00	11	00	00	00
S_g	00	00	00	00	00	00	00	00

$$x \cdot \overline{f} + \overline{g} \cdot f$$

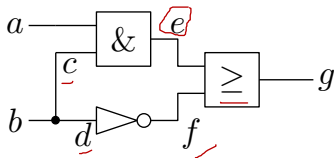
Exercise 1b) Fault Parallel Simulation with X

input vector
(a,b) = (1, X)



$$\begin{aligned}
 a &= \underline{(11\ 11\ 11\ 11\ 11\ 11\ 11\ 11)} \cdot I'_a + S_a \cdot I_a \\
 &= (11\ 11\ 11\ 11\ 11\ 11\ 11\ 11) \cdot (11\ 00\ 00\ 11\ 11\ 11\ 11\ 11) + \\
 &\quad (00\ 00\ \underline{11}\ 00\ 00\ 00\ 00\ 00) \cdot (00\ 11\ 11\ 00\ 00\ 00\ 00\ 00) \\
 &= \underline{11\ 00}\ 11\ 11\ 11\ 11\ 11\ 11. \\
 b &= (\underline{01}\ 01\ 01\ 01\ 01\ 01\ 01\ 01) \cdot I'_b + S_b \cdot I_b \\
 &= (01\ 01\ 01\ 01\ 01\ 01\ 01\ 01) \cdot (11\ 11\ 11\ 11\ 11\ 11\ 11\ 11) + \\
 &\quad (00\ 00\ 00\ 00\ 00\ 00\ 00\ 00) \cdot (00\ 00\ 00\ 00\ 00\ 00\ 00\ 00) \\
 &= 01\ 01\ 01\ 01\ 01\ 01\ 01\ 01.
 \end{aligned}$$

Exercise 1b) Fault Parallel Simulation with X



$$d = c = b; \quad \underline{e = (a \cdot c)} = (\underline{11} \underline{00} 11 11 11 11 11 11) \cdot (\underline{01} \underline{01} 01 01 01 01 01 01) =$$

$$\underline{01} \underline{00} 01 01 01 01 01 01.$$

$$f = (d') \cdot I'_f + I_f \cdot S_f$$

$$= (\underline{01} \underline{01} 01 01 01 01 01 01) \cdot (11 11 11 \underline{00} 11 11 11 11) + \dots$$

5-a-1 on f

$$= \underline{01} \underline{01} 01 \underline{11} 01 01 01 01.$$

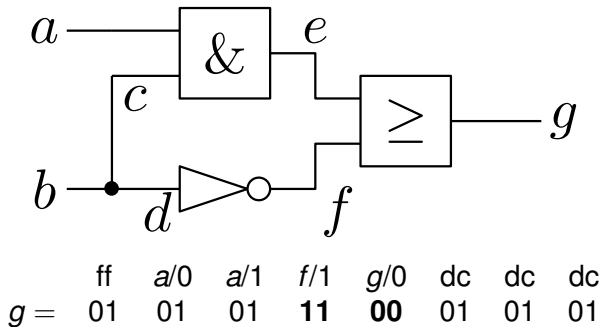
$$e + f = (01 00 01 01 01 01 01 01) + (01 01 01 11 01 01 01 01) = 01 01 01 11 01 01 01 01$$

$$g = (e + f) \cdot I'_g + I_g \cdot S_g$$

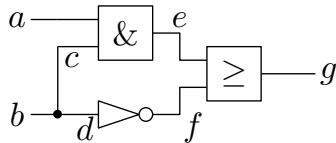
$$= (\underline{01} \underline{01} 01 \underline{11} 01 01 01 01) \cdot (11 11 11 11 00 11 11 11) + \dots$$

$$= \underline{01} \underline{01} 01 \underline{11} \underline{00} 01 01 01.$$

Exercise 1c) *Fault* Parallel Simulation with X



Exercise 1d) *Fault* Parallel Simulation with X, Pattern $(a, b) = (1, 0)$



$$a = (11\ 11\ 11\ 11\ 11\ 11\ 11\ 11\ 11) \cdot (11\ 00\ 00\ 11\ 11\ 11\ 11\ 11\ 11) + (00\ 00\ 11\ 00\ 00\ 00\ 00\ 00\ 00) \cdot (00\ 11\ 11\ 00\ 00\ 00\ 00\ 00\ 00) \\ = 11\ 00\ 11\ 11\ 11\ 11\ 11\ 11\ 11.$$

$$b = (00\ 00\ 00\ 00\ 00\ 00\ 00\ 00\ 00) \cdot (11\ 11\ 11\ 11\ 11\ 11\ 11\ 11\ 11) + (00\ 00\ 00\ 00\ 00\ 00\ 00\ 00\ 00) \cdot (00\ 00\ 00\ 00\ 00\ 00\ 00\ 00\ 00) \\ = 00\ 00\ 00\ 00\ 00\ 00\ 00\ 00\ 00.$$

$$d = c = b$$

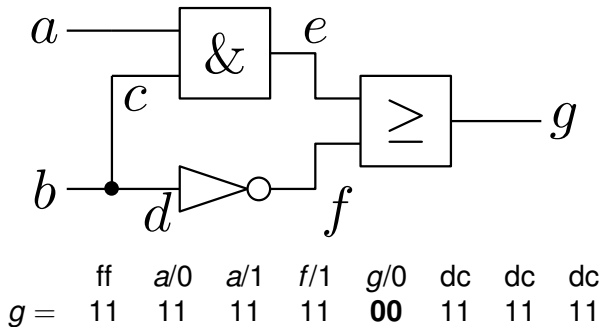
$$e = (a \cdot c) = (11\ 00\ 11\ 11\ 11\ 11\ 11\ 11\ 11) \cdot (00\ 00\ 00\ 00\ 00\ 00\ 00\ 00\ 00) = 00\ 00\ 00\ 00\ 00\ 00\ 00\ 00\ 00.$$

$$f = (11\ 11\ 11\ 11\ 11\ 11\ 11\ 11\ 11) \cdot (11\ 11\ 11\ 00\ 11\ 11\ 11\ 11\ 11) + (00\ 00\ 00\ 11\ 00\ 00\ 00\ 00\ 00) \cdot (00\ 00\ 00\ 11\ 00\ 00\ 00\ 00\ 00) \\ = 11\ 11\ 11\ 11\ 11\ 11\ 11\ 11\ 11.$$

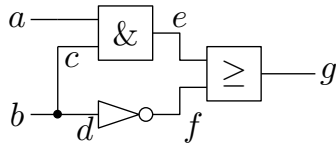
$$e + f = (00\ 00\ 00\ 00\ 00\ 00\ 00\ 00\ 00) + (11\ 11\ 11\ 11\ 11\ 11\ 11\ 11\ 11) = 11\ 11\ 11\ 11\ 11\ 11\ 11\ 11\ 11$$

$$g = (11\ 11\ 11\ 11\ 11\ 11\ 11\ 11\ 11) \cdot (11\ 11\ 11\ 11\ 00\ 11\ 11\ 11\ 11) + (00\ 00\ 00\ 00\ 11\ 00\ 00\ 00\ 00) \cdot (00\ 00\ 00\ 00\ 00\ 00\ 00\ 00\ 00) \\ = 11\ 11\ 11\ 11\ 11\ 00\ 11\ 11\ 11.$$

Exercise 1d) *Fault* Parallel Simulation with X, Pattern $(a, b) = (1, 0)$



Exercise 1d) *Fault* Parallel Simulation with X, Pattern $(a, b) = (1, 1)$



$$a = (11\ 11\ 11\ 11\ 11\ 11\ 11\ 11\ 11) \cdot (11\ 00\ 00\ 11\ 11\ 11\ 11\ 11\ 11) + (00\ 00\ 11\ 00\ 00\ 00\ 00\ 00\ 00) \cdot (00\ 11\ 11\ 00\ 00\ 00\ 00\ 00\ 00) \\ = 11\ 00\ 11\ 11\ 11\ 11\ 11\ 11\ 11.$$

$$b = (11\ 11\ 11\ 11\ 11\ 11\ 11\ 11\ 11) \cdot (11\ 11\ 11\ 11\ 11\ 11\ 11\ 11\ 11) + (00\ 00\ 00\ 00\ 00\ 00\ 00\ 00\ 00) \cdot (00\ 00\ 00\ 00\ 00\ 00\ 00\ 00\ 00) \\ = 11\ 11\ 11\ 11\ 11\ 11\ 11\ 11\ 11.$$

$$d = c = b$$

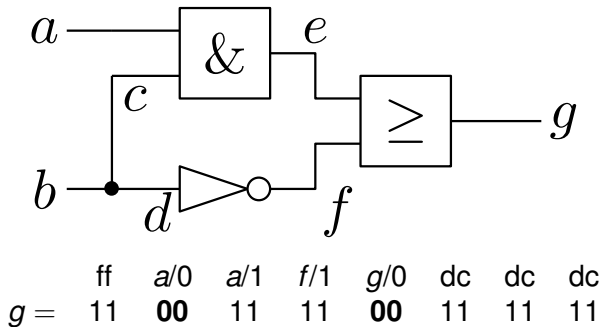
$$e = (a \cdot c) = (11\ 00\ 11\ 11\ 11\ 11\ 11\ 11\ 11) \cdot (11\ 11\ 11\ 11\ 11\ 11\ 11\ 11\ 11) = 11\ 00\ 11\ 11\ 11\ 11\ 11\ 11\ 11.$$

$$f = (00\ 00\ 00\ 00\ 00\ 00\ 00\ 00\ 00) \cdot (11\ 11\ 11\ 11\ 00\ 11\ 11\ 11\ 11) + (00\ 00\ 00\ 11\ 00\ 00\ 00\ 00\ 00) \cdot (00\ 00\ 00\ 11\ 00\ 00\ 00\ 00\ 00) \\ = 00\ 00\ 00\ 11\ 00\ 00\ 00\ 00\ 00.$$

$$e + f = (11\ 00\ 11\ 11\ 11\ 11\ 11\ 11\ 11) + (00\ 00\ 00\ 11\ 00\ 00\ 00\ 00\ 00) = 11\ 00\ 11\ 11\ 11\ 11\ 11\ 11\ 11$$

$$g = (11\ 00\ 11\ 11\ 11\ 11\ 11\ 11\ 11) \cdot (11\ 11\ 11\ 11\ 00\ 11\ 11\ 11\ 11) + (00\ 00\ 00\ 00\ 11\ 00\ 00\ 00\ 00) \cdot (00\ 00\ 00\ 00\ 00\ 00\ 00\ 00\ 00) \\ = \underline{11\ 00\ 11\ 11\ 00\ 11\ 11\ 11\ 11.}$$

Exercise 1d) *Fault* Parallel Simulation with X, Pattern $(a, b) = (1, 1)$



Exercise 2a), D-Algorithm

- All versions are complete
- Inaffected by order of propagation of D-/J-frontier
- D-frontier c is considered as harder in general, thus it is preferred in the original algorithm
- However, you can easily construct examples in which it might be favorable to prefer the J-frontier ...

Exercise 2b), D-Algorithm

- Controllability measures:
Indicate the difficulty of setting a line to a given value.
- Observability measures:
Indicate the difficulty of propagating an error from a line to a PO.

Exercise 2b), D-Algorithm



OR-gate g with n inputs $i_1, \dots, i_n$:

$$C0(g) = C0(i_1) + \dots + C0(i_n)$$

$$C1(g) = \min\{C1(i_1), \dots, C1(i_n)\}$$

$$O(i_1) = C0(i_2) + \dots + C0(i_n) + O(g) \text{ analogously for } i_2, \dots, i_n$$

NAND-gate g with n inputs $i_1, \dots, i_n$:

$$C0(g) = C1(i_1) + \dots + C1(i_n)$$

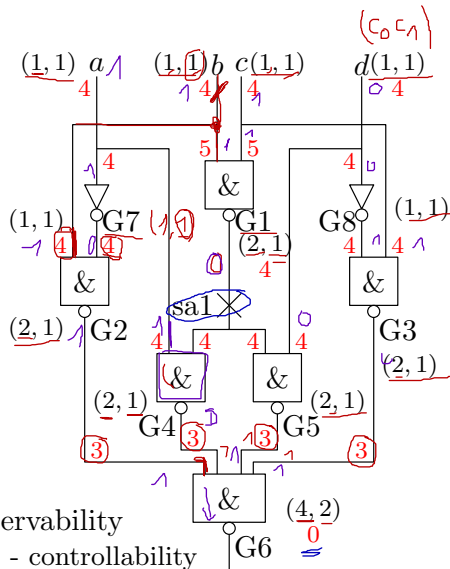
$$C1(g) = \min\{C0(i_1), \dots, C0(i_n)\}$$

$$O(i_1) = \underline{C1(i_2) + \dots + C1(i_n)} + \underline{O(g)} \text{ analogously for } i_2, \dots, i_n$$

Exercise 2c), D-Algorithm

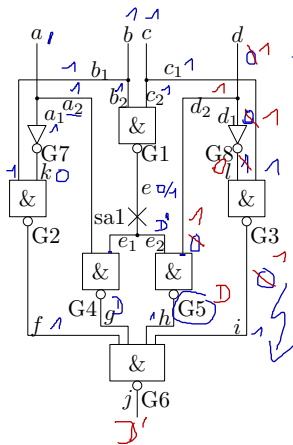
6-

D-fn. = {G4, G5}
G6



$\circ$ - observability
(C0,C1) - controllability

Exercise 2d), D-Alg.



t	Dec.	assignment list	D-Front	J-Fr.	Note
1		(e,D',f); (e,0,b)			fault activation
2		(b ₂ ,1,b); (c ₂ ,1,b); (e ₁ ,D',f); (e ₂ ,D',f)			ImPLY()
2		(b,1,b); (b ₁ ,1,f); (c,1,b); (c ₁ ,1,f)	{G4,G5}		ImPLY()
3	<u>G4</u>	(a ₂ ,1,b); (g,D,f)	{G5}		select
4		(a,1,b); (a ₁ ,1,f)	{G5,G6}		ImPLY()
4		(k,0,f)	{G5,G6}		ImPLY()
4		(f,1,f)	{G5,G6}		ImPLY()
5	<u>G6</u>	(f,1,b); (h,1,b); (j,1,b); (j,D',f)	{G5}		select
6		(d ₂ ,0,b); (l,0,b)			ImPLY()
6		(d ₁ ,0,f); (d,0,b); (d ₁ ,1,b)			ImPLY() conflict
7	G5	(d ₂ ,1,b); (h,D,f)	{G6}		backtrack and select
8		(d ₁ ,1,f); (d,1,b)	{G6}		ImPLY()
8		(l,0,f)	{G6}		ImPLY()
8		(i,1,f)	{G6}		ImPLY()
8		(j,D',f)			ImPLY()
9					error prop&D-front empty
10					lines just&J-front empty

Exercise 2e), D-Algorithm

Is it useful to utilize controllability and observability measures in order to find a test pattern for slow-to-fall transition delay faults?

Exercise 2f), PODEM

- Backtrace ($e, 0$): inversion is $i = 1$, we pick input b_2 of G_1 .
- Implications: $b = 1, b = 1, b_1 = 1$
- Backtrace ($e, 0$) again: implications $c_2 = 1, c = 1, c_1 = 1$
- D-frontier: $\{G_4, G_5\}$.
- G_4 : Backtrace ($a_2, 1$) : *implications* $a_2=1, a=1, a_1 = 1, k = 0, f = 1, g = D$.
- G_5 : Backtrace ($d_2, 1$): *implications* $d_2 = 1, d = 1, d_1 = 1, l = 0, i = 1, h = D, j = D'$.
- DONE.

Exercise 3a), Test Compaction (static)

Assume the test set:

$t_1 = 000x$

$t_2 = 101x$

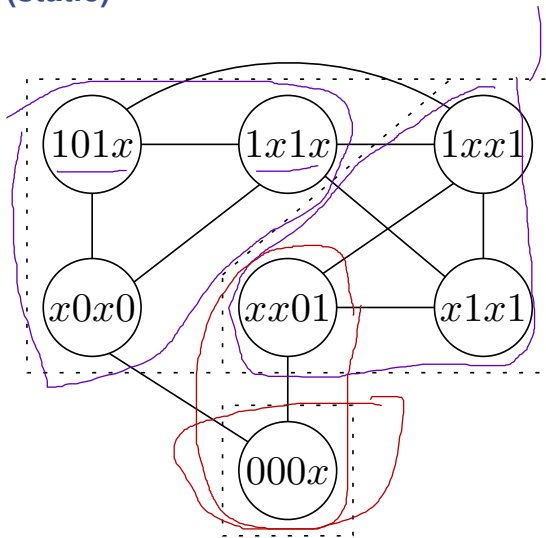
$t_3 = 1x1x$

$t_4 = 1xx1$

$t_5 = x0x0$

$t_6 = x1x1$

$t_7 = xx01$



Exercise 3b), Test Compaction (static)

Prove formally: All nodes of one clique in the compatibility graph can be compacted to one test pattern. In other words: For each clique $t_1, \dots, t_n$ there exists a test pattern $\tilde{t}$, which is compatible with t_i for each $i \in \{1, \dots, n\}$.

$n=1 \rightarrow \text{trivial}$

$n \rightarrow n+1 \rightarrow \text{remove 1 node } t_i$
 $\rightarrow \text{rest clique of size } n \rightarrow \text{ind. assumption} \rightarrow \tilde{t}^{\sim} \text{ comp. with all other nodes}$
 $t_i = \dots 1 \dots$
 $\tilde{t} = \dots \underline{0} \dots$
 $t_n = \dots 0$
 $\rightarrow t_i \text{ not compat to } t_n \quad \downarrow$

